

E072

Data for AI

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A retailer based in Austin with 900 employees collects a sales table with fixed columns, JSON clickstream events from its website, and audio recordings of support calls. Which source is semi-structured?

- A. All three, because they are stored by the company.
- B. The support call audio.
- C. The JSON clickstream events.
- D. The sales table.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A data scientist with a 50,000-row dataset standardizes every feature using the mean and standard deviation of the whole dataset, then splits 80/20 into training and test sets. The project is due in two weeks. What is wrong?

- A. Nothing; the whole dataset has the same distribution.
- B. The dataset is too small to split.
- C. Test data leaked into preprocessing; split first and fit the scaler on training data only.
- D. The scaler should be fit on the test set only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A fraud model scores 99% on its test set but about 70% in production. One feature is refund_issued, which is recorded days after the fraud decision. What should the team do?

- A. Remove features that are not available when the prediction is made.
- B. Collect more training data.
- C. Treat it as underfitting and add model complexity.
- D. Retrain with the same features for more epochs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An income column, pulled from a partner's CRM export, is strongly right-skewed by a few very high earners, and 8% of values are missing at random. The export runs nightly at 2 a.m. What is a sound way to fill the gaps?

- A. Impute the median, optionally with a missing-value flag.
- B. Impute the mean.
- C. Fill every blank with 0.
- D. Drop every row that has any missing value.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Heart-rate readings from 12 wearable devices range from 40 to 100 beats per minute. Using min-max scaling, what is the scaled value of 70?

- A. 0.3
- B. 0.5
- C. 0.7
- D. About 0.43

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A feature has mean 70 and standard deviation 10. A record's z-score is 1.5. What is its original value?

- A. 15
- B. 71.5
- C. 55
- D. 85

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A delivery model uses a city column with five values (Boise, Reno, Fresno, Tulsa, Omaha). The data engineer, who prefers Python over R, encodes them as 1-5 in alphabetical order. What is the problem and fix?

- A. No problem; any integers work for categories.
- B. Numbers invent an order the model may use; use one-hot encoding.
- C. Rewrite the pipeline in R.
- D. Drop the column to avoid encoding.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A shirt-size column has values S, M, L, and XL. Which encoding keeps the useful information?

- A. Apply min-max scaling to the letters.
- B. One-hot encoding is always required for text.
- C. Assign random integers to each size.
- D. Ordinal encoding: S=0, M=1, L=2, XL=3.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A test set has 1,000 transactions, 20 of them fraud. A model trained in 4 minutes on a GPU predicts every transaction as legitimate. How should its performance be described?

- A. 98% accuracy, so it is a strong model.
- B. 2% accuracy.
- C. Good, because it trained quickly.
- D. 98% accuracy but 0% recall on fraud — useless for catching fraud.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Of 20 actual fraud cases, a model flags 50 transactions, and 15 of those are real fraud. What is its recall on fraud?

- A. 75%
- B. 40%
- C. 30%
- D. 25%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

After a CRM was migrated from an on-premises server, 3% of customer records appear twice with the same customer ID. The migration took one weekend. Which data quality dimension is failing?

- A. Accuracy, because every data problem is an accuracy problem
- B. Uniqueness
- C. Completeness
- D. Timeliness

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A support team wants an AI assistant to find past help articles that mean the same thing as a customer’s question, even with different wording. There are 2 million articles written over ten years. Which data approach fits?

- A. Store the articles as files with no index.
- B. Use SQL LIKE keyword matching.
- C. Create embeddings of the articles and store them in a vector index for similarity search.
- D. Normalize the article tables to third normal form.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A retailer ~~based in Austin with 900 employees~~ collects a sales table with fixed columns, JSON clickstream events from its website, and audio recordings of support calls. Which source is semi-structured?

Target word: identify semi-structured data **Decides:** JSON carries its own keys without a fixed table schema.

A	Absolute language	Storage location does not define structure.
B	Confusion set	Audio is unstructured.
C	Correct	Self-describing keys, flexible structure.
D	Adjacent category	Fixed rows and columns are structured.

2. Answer C Concept 2

A data scientist ~~with a 50,000 row dataset~~ standardizes every feature using the mean and standard deviation of the whole dataset, then splits 80/20 into training and test sets. ~~The project is due in two weeks.~~ What is wrong?

Target word: spot the leakage **Decides:** Test data influenced the scaling statistics.

A	Confusion set	Even small leaks inflate test scores.
B	Camouflage	50,000 rows is plenty; size is not the problem.
C	Correct	Keeps the test set truly unseen.
D	Inverted operation	Moves the leak the other way.

3. Answer A Concept 2 · transfer

A fraud model scores 99% on its test set but about 70% in production. One feature is refund_issued, which is recorded days after the fraud decision. What should the team do?

Target word: fix target leakage **Decides:** The feature is not available at prediction time.

A	Correct	Eliminates the leak that inflated test results.
B	Adjacent category	More leaky data keeps the same leak.
C	Confusion set	The test score is high; this is leakage, not underfitting.
D	Dropped element	Leaves the leaking feature in.

4. Answer A Concept 3

An income column, ~~pulled from a partner's CRM export~~, is strongly right-skewed by a few very high earners, and 8% of values are missing at random. ~~The export runs nightly at 2 a.m.~~ What is a sound way to fill the gaps?

Target word: imputation choice for skewed data **Decides:** The column is skewed by outliers.

A	Correct	The median resists outliers.
B	Confusion set	High earners pull the mean up.
C	Adjacent category	Zero income is a real value and distorts the data.
D	Absolute language	Throws away 8% of rows when a simple fill works.

5. Answer B Concept 4

Heart-rate readings from 12 wearable devices range from 40 to 100 beats per minute. Using min-max scaling, what is the scaled value of 70?

Target word: min-max value **Decides:** $(70 - 40)/(100 - 40)$.

A	Adjacent category	Divides by the max, not the range.
B	Correct	$30 / 60 = 0.5$.
C	Dropped element	$70 / 100$ ignores the minimum.
D	Confusion set	$30 / 70$ divides by the value itself.

6. Answer D Concept 4 · transfer

A feature has mean 70 and standard deviation 10. A record's z-score is 1.5. What is its original value?

Target word: reverse a z-score **Decides:** $x = \text{mean} + z \times \text{SD}$.

A	Adjacent category	The distance from the mean, not the value.
B	Dropped element	Adds z without multiplying by the SD.
C	Inverted operation	Subtracts instead of adds.
D	Correct	$70 + 1.5 \times 10 = 85$.

7. Answer B Concept 5

A delivery model uses a city column with five values (Boise, Reno, Fresno, Tulsa, Omaha). The data engineer, who prefers Python over R, encodes them as 1-5 in alphabetical order. What is the problem and fix?

Target word: encoding unordered categories **Decides:** Cities have no natural order.

A	Confusion set	Integer codes imply order and distance.
B	Correct	One column per city with no implied ranking.
C	Camouflage	Language choice does not fix encoding.
D	Dropped element	Loses possibly useful information.

8. Answer D Concept 5 · transfer

A shirt-size column has values S, M, L, and XL. Which encoding keeps the useful information?

Target word: encoding ordered categories **Decides:** Sizes have a real order.

A	Adjacent category	Scaling applies to numbers, not labels.
B	Absolute language	Works, but throws away the order for ordered data.
C	Inverted operation	Scrambles the real order.
D	Correct	Preserves the real order.

9. Answer D Concept 6

A test set has 1,000 transactions, 20 of them fraud. A model trained in 4 minutes on a GPU predicts every transaction as legitimate. How should its performance be described?

Target word: evaluate on imbalanced data **Decides:** It catches none of the 20 fraud cases.

A	Confusion set	Accuracy hides the rare-class failure.
B	Inverted operation	That is the fraud rate, not accuracy.
C	Camouflage	Training speed does not measure quality.
D	Correct	980/1,000 correct, but no fraud found.

10. Answer A Concept 6 · transfer

Of 20 actual fraud cases, a model flags 50 transactions, and 15 of those are real fraud. What is its recall on fraud?

Target word: compute recall **Decides:** Recall = true positives / actual positives.

A	Correct	15 / 20.
B	Adjacent category	20 / 50 divides actual fraud by flags, which is neither recall nor precision.
C	Confusion set	15 / 50 is precision.
D	Inverted operation	5 / 20 is the miss rate.

11. Answer B Concept 7

After a CRM was migrated ~~from an on-premises server~~, 3% of customer records appear twice with the same customer ID. ~~The migration took one weekend.~~ Which data quality dimension is failing?

Target word: name the dimension **Decides:** The same entity appears twice.

A	Absolute language	Quality has distinct dimensions; this symptom is duplication.
B	Correct	Duplicate records break uniqueness.
C	Confusion set	Completeness is about missing values.
D	Adjacent category	About how current data is.

12. Answer C Concept 8

A support team wants an AI assistant to find past help articles that mean the same thing as a customer's question, even with different wording. There are 2 million articles ~~written over ten years~~. Which data approach fits?

Target word: meaning-based retrieval **Decides:** Match by meaning, not exact words.

A	Dropped element	No way to search by meaning.
B	Adjacent category	Finds exact words, misses paraphrases.
C	Correct	Nearby vectors capture similar meaning.
D	Confusion set	Reduces redundancy; does not enable meaning search.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Structured, semi-structured, unstructured	Structured data fits fixed rows and columns; semi-structured carries its own tags or keys (JSON, XML, logs); unstructured has no predefined model (text, audio, images).	Calling data structured just because it sits in a database or file store.	The format of the data, not where it is stored.
2	Data leakage	Split data first and fit every preprocessing step on training data only; never use features that would not exist at prediction time.	Scaling or imputing on the full dataset, or including a field recorded after the outcome.	Test scores that are too good to be true, or a big drop in production.
3	Handling missing values	Choose imputation to fit the data: median for skewed numbers, mode for categories, and consider a missing-indicator flag; drop rows only when few and random.	Using the mean on skewed data, or dropping every row with any gap.	A skewed column with a modest share of blanks.
4	Scaling features	Min-max scaling = $(x - \min) / (\max - \min)$ into 0-1; z-score = $(x - \text{mean}) / \text{standard deviation}$.	Dividing by the max instead of the range, or forgetting to multiply back by the standard deviation.	A value with a min and max, or a mean and standard deviation.
5	Encoding categorical data	Use one-hot encoding for categories with no order; use ordinal encoding only when the categories have a real order.	Numbering unordered categories 1, 2, 3, which invents an order.	Names like cities or colors vs. levels like S/M/L.
6	Class imbalance and metrics	With a rare class, accuracy can look high while the model misses every case; judge with recall and precision for the rare class.	Trusting 98% accuracy on a 2% fraud rate.	One class is a small fraction of the data.
7	Data quality dimensions	Completeness (nothing missing), uniqueness (no duplicates), validity (fits rules and formats), consistency (agrees across sources), timeliness (current), accuracy (true to reality).	Calling every data problem an accuracy problem.	The specific symptom: blanks, duplicates, bad formats, stale values.
8	Embeddings and vector databases	Unstructured content can be turned into embedding vectors and stored in a vector index so AI systems can search by meaning (similarity), not exact words.	Trying to do meaning-based search with keyword matching or table normalization.	“Find similar” or “search by meaning” over text or images.

Course scope (public description): This course examines data through a modern artificial intelligence (AI) lens, focusing on how data is sourced, prepared, and validated to support AI-driven systems. Learners explore the role of databases and data architectures in AI, including structured and unstructured data, and examine algorithms and techniques used to enable intelligent applications. Emphasis is placed on preparing, transforming, and validating

data for use in AI systems. By the end of the course, learners will be able to support modern datadriven applications by applying data preparation, transformation, and testing techniques to ensure data quality and reliability for AI applications.